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NEPAL

STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Nepal is assessed to be at low risk of debt distress compared to the previous assessment of moderate risk, mainly due to reduced estimates of the cost of a potential financial sector shock and the increase in the discount rate used.[1] Generally prudent fiscal policy and low execution of capital spending budgets have underpinned declining levels of public debt. The baseline external public debt indicators show that external debt dynamics are broadly sound and remain resilient to standard stress tests.[2] Efforts to raise capital investment would enhance long term growth; in this context, raising net incurrence of liabilities to around 2½ percent of GDP over the medium term would help boost domestic capacity while maintaining a stable debt profile.

BACKGROUND

1. The total stock of public debt in Nepal declined in 2013 to 31¼ percent of GDP from 36 percent in 2012, and from around 60 percent a decade ago, largely reflecting prudent fiscal policy. External debt stood at 20 percent of GDP by end-2013 ($3½ billion), of which 85 percent is concessional borrowing from World Bank and Asian Development Bank (ADB). Japan is the largest bilateral creditor, followed by Korea, India and China. Domestic debt declined from 13½ percent of GDP in 2012 to 11½ percent by end-2013, on account of low budget execution resulting in a budget surplus in 2012/13.

[1] The risk rating is determined using the Low Income Country Debt Sustainability Analysis (LIC-DSA) framework. Nepal's fiscal year starts in mid-July.

[2] The thresholds are determined based on Nepal's policy performance rating, which is "medium" according to the CPIA score.

MACROECONOMIC ASSUMPTIONS

2. Macroeconomic assumptions remain broadly in line with the previous DSA (Box 1). Key differences, reflecting developments since the previous DSA, include (i) moderately improved growth prospects, (ii) higher revenue gains over the long term, based on better-than-anticipated performance to date and in line with continued emphasis on revenue administration reforms, and (iii) a slightly stronger current account in the near and medium term, driven by higher-than-anticipated remittance growth, which is only partially offset by stronger imports. In the long term, however, the current account is expected to be slightly weaker.

- Real GDP growth is expected to recover from 3½ percent in 2012/13 to 4¾ percent in 2013/14, mainly driven by a recovery in agricultural output, a buoyant services sector supported by remittances, and higher public spending. The election of a new Constituent Assembly and government has reduced the degree of political uncertainty. The baseline assumes improved budget execution of capital spending compared to the previous DSA in line with authorities' efforts in this area, as growth of remittance inflows moderates. Overall, growth in the medium and long run is projected at 4½ percent, slightly higher than 4 percent assumed previously.

- The external current account is projected to move from a sizeable surplus in 2013/14 to moderate deficits over the medium to long term. The exchange rate peg with the Indian rupee is assumed to remain at the current level over the projection period. Import growth is expected to moderate in line with remittances. Export growth is projected to increase only

Box 1. Macroeconomic Assumptions Table

	Previous DSA				Current DSA			Current versus previous	
	2012	**2014**	**MT**	**LT**	**2014**	**MT**	**LT**	**MT**	**LT**
Real growth (%)	4.6	3.9	4.0	4.0	4.8	4.7	4.5	0.7	0.5
Inflation (GDP deflator, %)	8.7	8.1	7.7	5.1	8.8	7.1	5.0	-0.6	-0.1
Revenues and grants (% GDP)	18.3	17.9	18.1	19.2	21.1	21.7	22.5	3.6	3.3
Grants (% GDP)	2.6	2.0	2.1	1.8	2.8	2.5	2.3	0.4	0.5
Primary expenditure (% GDP)	18.7	19.2	19.3	21.3	21.4	22.9	24.0	3.6	2.7
Net acquisition of non-financial assets (% GDP)	3.1	3.3	3.3	4.2	3.7	4.4	5.2	1.1	1.0
Primary deficit (% GDP)	0.4	-0.6	1.2	2.1	0.3	1.1	1.5	-0.1	-0.6
Net incurrence of liabilities	1.4	2.3	2.5	3.6	1.3	2.1	2.5	-0.4	-1.1
Net domestic financing (% GDP)	1.4	2.3	2.4	2.7	1.0	1.0	1.5	-1.4	-1.2
Exports of G&S (y/y growth)	11.7	8.2	7.7	7.1	4.1	7.2	6.0	-0.5	-1.1
Imports of G&S (y/y growth)	0.6	9.4	9.3	7.4	10.9	10.2	6.1	0.9	-1.3
Remittances (y/y growth)	24.5	8.3	10.9	7.5	15.0	7.9	6.0	-3.0	-1.5
Current account balance (% GDP)	4.7	-0.3	-0.5	-0.5	4.2	0.9	-0.8	1.4	-0.3

Note: MT stands for medium term and reflects average over the next 5 years, and LT refers to long term and generally reflcts indicators at the end of the projection period.

moderately, reflecting weak competitiveness due to significant infrastructure bottlenecks. As a consequence, the ratio of exports to GDP is expected to gradually decline over the medium term.

- Fiscal policy is expected to remain prudent. Enhanced revenue generation backed by tax administration reforms, augmented external grants (which, though, decline over time as a share of GDP), and moderately higher fiscal deficits would allow for higher public investment, which would increase from 3.7 percent of GDP in 2014 to 5.2 percent in 2034. Net incurrence of liabilities is projected to rise from 1¼ percent of GDP in 2013/14 to 2¼ percent over the next five years, rising gradually to 2½ percent towards the end of the DSA horizon. This path is consistent with a stable debt profile. Financing of the deficit is expected to tilt increasingly towards domestic sources (rising to 1½ percent of GDP in the long term), as public financial management improves and external loans decline relative to GDP.

- The grant element of foreign borrowing has increased relative to the previous DSA, mainly on account of the upward revision of the discount rate to 5 percent. The change in the discount rate, unified at 5 percent for all LIC DSAs, was made to prevent a narrowing of borrowing space for LICs due to historically low advanced economy interest rates and inflated debt burden estimates. As a result of the higher discount rate, the grant element remains in the range of 53–55 percent throughout the DSA horizon compared to 32–36 percent in the previous DSA, and one breach observed in the previous DSA (PV of external debt to exports + remittances in response to the most extreme shock) is no longer observed. Future debt disbursements are assumed to be mostly concessional IDA and ADB loans, with enhancement of total IDA assistance by at least 10 percent, triggered by the low-risk assessment.[3]

EXTERNAL DEBT SUSTAINABILITY

A. Baseline

3. Under the baseline scenario, Nepal's external debt indicators remain well below indicative sustainability thresholds (Figure 1 and, Table 3b). As in the previous DSA, remittances are formally included in the analysis as inflows remain robust, and are poised to touch nearly 30 percent of GDP in 2013/14. However, debt dynamics may be susceptible to volatility in remittance flows, as captured under standard shocks, discussed below. Over the medium term, the present value (PV) of external debt stabilizes at 8 percent of GDP + remittances, 25 percent of exports + remittances,

[3] The change in risk rating from "moderate" to "low" would lead to a switch from a 45–55 percent mix of grants and loans, respectively, from IDA to 100 percent financing on standard IDA credit terms, with a roughly 10 percent increase in the level of total assistance. This has been incorporated into the projections. The ADB already provides 100 percent loan assistance.

and 50 percent of revenues. Debt service-to-exports + remittances stabilizes at 1½ percent, while the ratio of debt service to revenues stabilizes at 3¼ percent.

B. Stress Tests and Alternative Scenarios

4. Debt dynamics remain resilient to standard shocks. These stress tests include shocks to GDP growth, exports, non-debt creating flows, and a combination of these shocks, as well as a one-time 30 percent nominal depreciation shock. Under the most severe shock (to non-debt creating flows, capturing a remittance shock), the PV of debt to exports + remittances rises rapidly over the next 2 years but stays below the threshold, and thereafter declines again, while all other indicators remain well below the thresholds.

PUBLIC DEBT SUSTAINABILITY

5. Under the baseline, the ratio of public debt to GDP rises modestly by the end of the projection period. The public debt-to-GDP ratio increases slightly from 31¼ percent in 2013 to 32½ percent in 2034. In PV terms, public debt to GDP declines somewhat from 25½ percent in 2013 to 23¾ percent by 2034. As a ratio of revenues and grants, the PV of public debt declines from 132 percent in 2013 to 103 percent. The composition of public debt is projected to tilt somewhat towards higher domestic debt, from 36 percent in 2013 to 40 percent of total public debt in 2034.

6. Debt dynamics remain resilient under standard stress tests. In the context of the PV of public debt-to-GDP ratio, the most extreme shock is a 30 percent one-time depreciation in 2015, which does not lead to a breach of the threshold. Among the other stress tests, the 10 percent of GDP increase in debt creating flows shock mimics a financial sector shock leading to domestic debt-financed bank recapitalization needs.[4] Under this shock, the PV of public debt-to-GDP ratio rises to a peak of 29 percent in 2015, again staying well below the 56 percent threshold. The financial sector shock assumed in the previous DSA, which led to sustained breaches on multiple debt indicators, is considered unlikely in view of the stress test findings in the recent assessment under the Financial Sector Assessment Program, and some progress on financial sector reforms.[5]

7. Contingent liabilities arise mainly from the operations of state owned enterprises (SOEs), and rising pension costs need to be addressed to head off future risks. Nepal Oil Corporation (NOC) and Nepal Electricity Authority (NEA) are the two biggest loss-making SOEs, on average making combined losses of 1½ percent of GDP a year, and needing frequent government bail-outs despite periodic (though not automatic) price adjustments to recover costs. In addition,

[4] A stress test conducted during the recent FSAP suggests that the recapitalization needs for the entire banking sector (public and private) from a financial sector shock in which nonperforming assets increase by 2 percentage points (implying a 50 percent increase) could amount to 4¾ percent of GDP. This test, based on updated calibration, supersedes the one conducted in the previous DSA.

[5] Key assumptions included (i) 50 percent loss of central bank foreign exchange reserves; (ii) a one-time 33 percent depreciation of the currency; (iii) an output loss of 30 percent over 4 years; and (iv) fiscal costs of bank resolution and deposit coverage of 23 percent of GDP financed through domestic and foreign debt.

NOC's outstanding loans (arising from accumulated past losses) alone stand at nearly NR 34 billion, or 1¾ percent of GDP. NEA's outstanding loans from the Government stood at nearly NR 69 billion (3½ percent of GDP). Other contingent liabilities of the SOE sector, including debt arrears, accumulated losses, government guaranteed debt, and unfunded liabilities to employees, amount to another 2 percent of GDP. Government guaranteed debt, mainly loans from the pension fund to NOC, is estimated at ¾ percent of GDP, and does not appear to pose a risk to debt sustainability. Civil service pension liabilities, are currently modest at 1¼ percent of GDP, rising to 1½ percent by 2025, and can be addressed through adequate parametric reforms in the medium term according to a recent IMF TA mission on pension reforms.

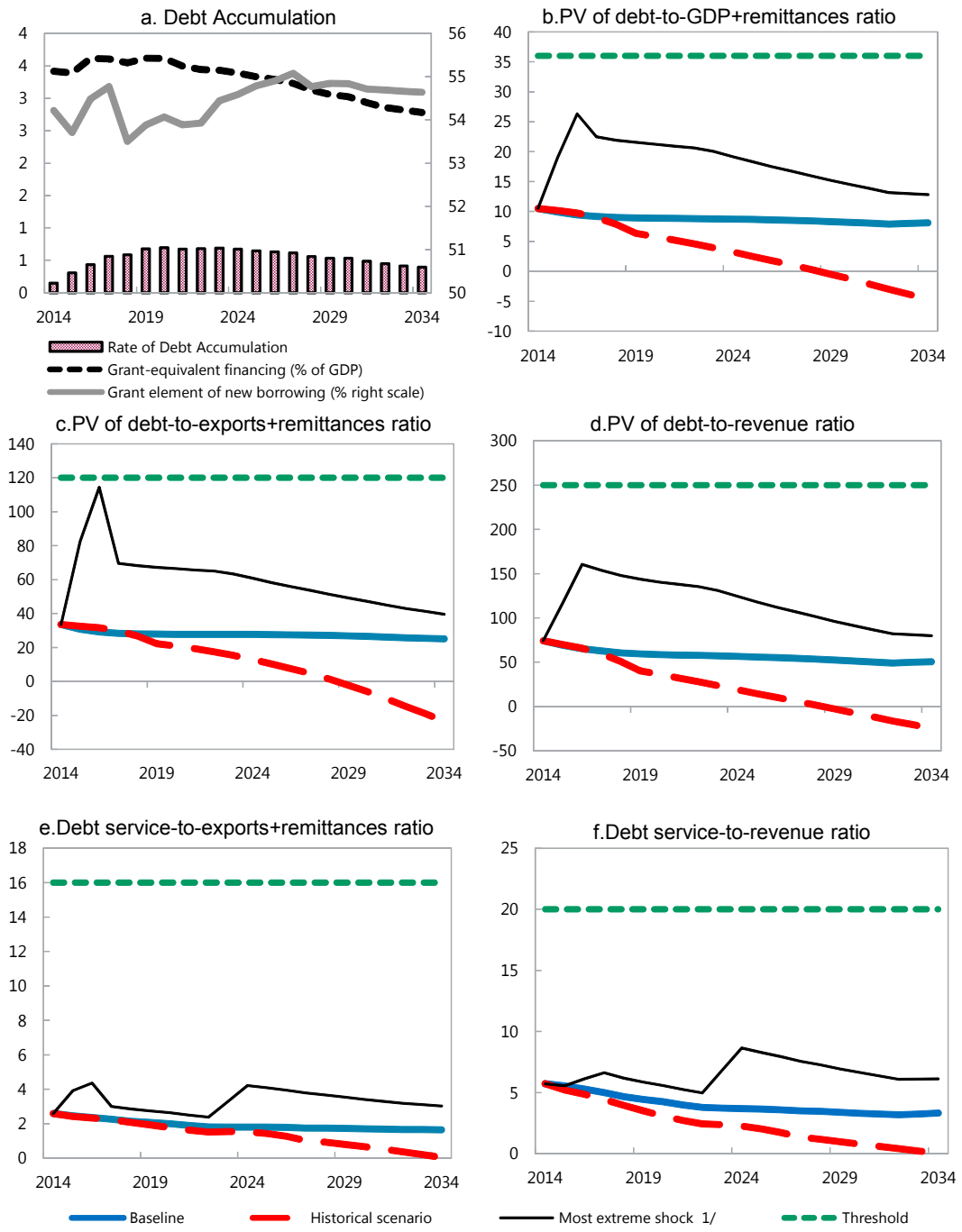
AUTHORITIES' VIEWS

8. The authorities broadly concurred with the findings and policy messages of the DSA. They expressed some concern over the change in IDA assistance from a mix of loans and grants to loans only triggered by the assessment of a low risk of debt distress, and pointed out that they were being penalized for their success. With respect to improving execution of capital spending, they highlighted various measures that they intend to implement to improve the situation. These include introducing the budget a month earlier in FY2015 to speed up the release of budget appropriations for investment projects, as well as amendments to the Procurement Act, and the drafting of a Fiscal Responsibility and Budget Management Law to enhance the budgeting and expenditure process.

CONCLUSION

9. The DSA suggests Nepal's risk of debt distress is low. This changes the assessment of moderate risk of debt distress in the previous DSA. The moderately improved medium term growth path and the higher discount factor underlying the current DSA imply a lower PV of public debt path than the previous assessment, even with a moderate increase in the size of fiscal deficits to enable higher capital spending in the medium term. The change implies that the mix of IDA assistance to Nepal will change to 100 percent IDA loans, but the level of IDA assistance will also increase by about 10 percent since the 20 percent discount applied to grants will not apply to a fully loan-based allocation. In order to maximize the growth benefits of somewhat enhanced allocation and the space afforded by the low debt trajectory, Nepal should make efforts to improve public financial management to boost public investment, by enhancing project appraisal, streamlining the budget release process, and strengthening monitoring of capital budget execution.

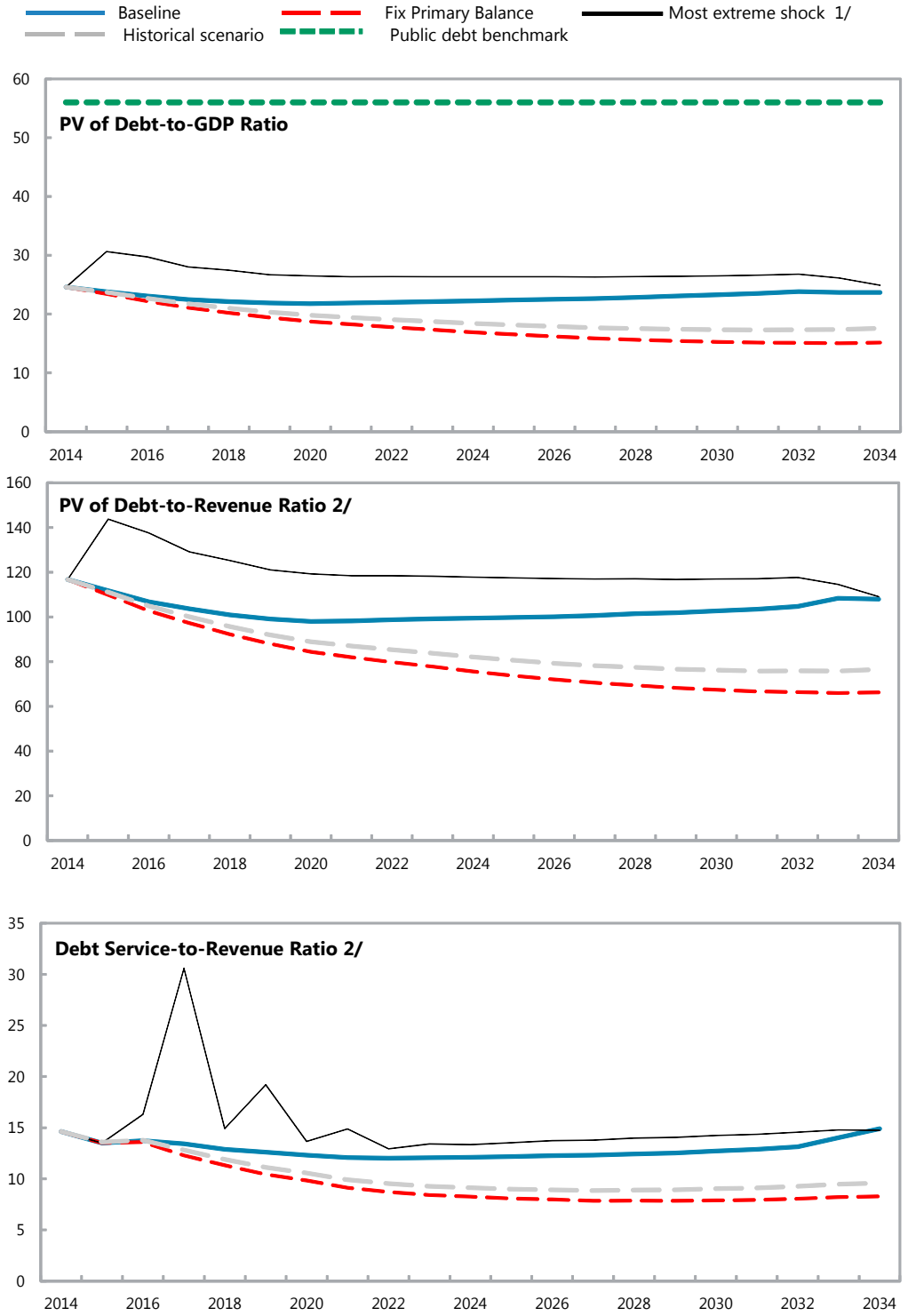
Figure 1. Nepal: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014–2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a Non-debt flows shock; in e. to a Non-debt flows shock and in figure f. to a Non-debt flows shock

Figure 2. Nepal: Indicators of Public Debt Under Alternative Scenarios, 2014–2034 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 1a. Nepal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–2034

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	34.5	36.3	31.5			30.5	29.7	29.1	28.8	28.7	28.8		30.4	32.5	
of which: foreign-currency denominated	19.9	22.7	20.1			19.5	18.9	18.4	18.4	18.4	18.6		19.4	19.4	
Change in public sector debt	-2.8	1.7	-4.8			-1.0	-0.8	-0.6	-0.3	-0.1	0.1		0.3	0.3	
Identified debt-creating flows	-3.7	3.0	-2.6			-1.0	-0.8	-0.6	-0.3	-0.1	0.1		0.3	0.1	
Primary deficit	1.1	1.2	-1.4	0.5	1.0	0.3	0.9	1.1	1.2	1.2	1.3	1.0	1.6	1.3	1.5
Revenue and grants	17.7	18.7	19.3			21.1	21.3	21.6	21.7	21.9	22.1		22.4	23.0	
of which: grants	3.3	2.7	1.8			2.8	2.6	2.6	2.5	2.5	2.5		2.4	2.1	
Primary (noninterest) expenditure	18.8	19.9	17.9			21.4	22.2	22.7	22.9	23.2	23.4		23.9	24.3	
Automatic debt dynamics	-4.8	1.8	-1.2			-1.4	-1.7	-1.7	-1.5	-1.4	-1.3		-1.3	-1.2	
Contribution from interest rate/growth differential	-2.0	-1.8	-1.6			-1.7	-1.7	-1.5	-1.3	-1.3	-0.9		-1.3	-1.2	
of which: contribution from average real interest rate	-0.7	-0.2	-0.2			-0.3	-0.3	-0.1	-0.1	0.0	0.4		0.0	0.2	
of which: contribution from real GDP growth	-1.2	-1.6	-1.3			-1.5	-1.4	-1.3	-1.3	-1.3	-1.2		-1.3	-1.4	
Contribution from real exchange rate depreciation	-2.8	3.6	0.4			0.4	0.0	-0.2	-0.1	-0.1	-0.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.0	-1.3	-2.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	25.5			24.6	23.8	23.1	22.5	22.1	21.9		22.2	23.7	
of which: foreign-currency denominated	...	...	14.1			13.6	13.0	12.4	12.1	11.8	11.7		11.3	10.6	
of which: external	...	...	14.1			13.6	13.0	12.4	12.1	11.8	11.7		11.3	10.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.4	6.2	4.5			4.4	4.8	5.0	5.1	5.0	5.1		5.2	6.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	132.2			116.7	111.7	106.8	103.7	100.9	99.0		99.5	102.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	145.8			134.4	127.0	121.7	117.2	114.0	111.7		111.3	113.0	
of which: external 3/	...	...	80.5			74.3	69.2	65.4	62.8	60.7	59.5		56.7	50.7	
Debt service-to-revenue and grants ratio (in percent) 4/	14.5	12.4	15.0			14.6	13.5	13.7	13.4	12.9	12.6		12.1	14.9	
Debt service-to-revenue ratio (in percent) 4/	17.8	14.5	16.5			16.9	15.4	15.7	15.2	14.6	14.2		13.6	16.4	
Primary deficit that stabilizes the debt-to-GDP ratio	3.8	-0.5	3.4			1.3	1.7	1.7	1.4	1.4	1.2		1.2	1.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.4	4.8	3.9	4.2	0.9	4.8	5.0	4.7	4.5	4.5	4.5	4.7	4.5	4.5	4.5
Average nominal interest rate on forex debt (in percent)	0.9	1.0	1.9	1.1	0.3	1.0	1.0	1.1	1.2	1.2	1.2	1.1	1.0	1.0	1.0
Average real interest rate on domestic debt (in percent)	-3.9	-0.4	-2.8	-3.1	3.2	-1.3	-0.6	0.3	1.0	1.3	1.7	0.4	2.2	3.0	2.3
Real exchange rate depreciation (in percent, + indicates depreciation)	-12.3	19.1	1.8	-3.2	10.6	1.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	10.8	6.6	6.7	8.6	4.0	8.8	7.8	7.0	6.3	6.0	5.5	6.9	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.1	-0.1	0.1	0.1	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	54.2	53.7	54.5	54.8	53.5	53.9	54.1	54.6	54.6	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 1b. Nepal: Sensitivity Analysis for Key Indicators of Public Debt, 2014–2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	25	24	23	22	22	22	22	24
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	24	23	22	21	20	18	18
A2. Primary balance is unchanged from 2014	25	23	22	21	20	19	17	15
A3. Permanently lower GDP growth 1/	25	24	23	23	23	22	24	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	25	24	24	24	24	24	26	28
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	25	24	24	23	23	22	23	23
B3. Combination of B1-B2 using one half standard deviation shocks	25	24	24	23	23	23	24	26
B4. One-time 30 percent real depreciation in 2015	25	30	28	27	26	25	24	23
B5. 10 percent of GDP increase in other debt-creating flows in 2015	25	31	30	28	27	27	26	25
PV of Debt-to-Revenue Ratio 2/								
Baseline	117	112	107	104	101	99	99	108
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	117	111	105	100	96	92	82	77
A2. Primary balance is unchanged from 2014	117	110	103	97	92	88	76	66
A3. Permanently lower GDP growth 1/	117	112	107	105	103	101	106	118
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	117	114	112	110	109	108	115	121
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	117	114	110	107	104	102	102	99
B3. Combination of B1-B2 using one half standard deviation shocks	117	113	109	107	105	104	109	113
B4. One-time 30 percent real depreciation in 2015	117	139	130	123	118	113	105	99
B5. 10 percent of GDP increase in other debt-creating flows in 2015	117	144	138	129	125	121	118	109
Debt Service-to-Revenue Ratio 2/								
Baseline	15	14	14	13	13	13	12	15
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	15	14	14	13	12	11	9	10
A2. Primary balance is unchanged from 2014	15	14	14	12	11	10	8	8
A3. Permanently lower GDP growth 1/	15	14	14	14	13	13	13	16
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	15	14	14	14	14	14	14	17
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	15	14	14	15	14	13	12	14
B3. Combination of B1-B2 using one half standard deviation shocks	15	14	14	14	13	14	14	16
B4. One-time 30 percent real depreciation in 2015	15	15	16	16	16	16	15	16
B5. 10 percent of GDP increase in other debt-creating flows in 2015	15	14	16	31	15	19	13	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 2a. Nepal: External Debt Sustainability Framework, Baseline Scenario, 2011–2034 1/

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**20.1**	**22.9**	**20.2**			**19.5**	**18.9**	**18.4**	**18.4**	**18.4**	**18.6**		**19.4**	**19.4**	
of which: public and publicly guaranteed (PPG)	19.9	22.7	20.1			19.5	18.9	18.4	18.4	18.4	18.6		19.4	19.4	
Change in external debt	-3.9	2.8	-2.6			-0.7	-0.6	-0.4	0.0	0.0	0.2		0.1	0.5	
Identified net debt-creating flows	-3.3	-5.2	-4.3			-5.7	-4.9	-3.7	-2.4	-1.1	0.1		-1.3	0.0	
Non-interest current account deficit	**0.8**	**-5.0**	**-3.5**	**-2.1**	**2.3**	**-4.3**	**-3.5**	**-2.4**	**-1.1**	**0.3**	**1.4**		**0.1**	**1.4**	0.5
Deficit in balance of goods and services	24.1	23.5	26.8			30.4	32.8	34.1	35.3	36.1	36.9		34.8	37.7	
Exports	9.0	10.1	10.7			11.1	11.2	11.1	11.1	11.2	11.2		11.1	11.3	
Imports	33.1	33.6	37.5			41.5	43.9	45.2	46.4	47.2	48.1		45.9	49.0	
Net current transfers (negative = inflow)	-22.6	-27.5	-29.4	-22.0	4.5	-33.7	-35.2	-35.5	-35.4	-34.9	-34.6		-34.1	-35.8	-34.0
of which: official	-1.7	-2.1	-1.3			-1.9	-1.8	-1.8	-1.7	-1.6	-1.6		-1.9	-2.4	
Other current account flows (negative = net inflow)	-0.7	-1.0	-0.9			-1.0	-1.0	-1.0	-1.0	-0.9	-0.9		-0.7	-0.5	
Net FDI (negative = inflow)	**-0.5**	**-0.6**	**-0.5**	**-0.2**	**0.2**	**-0.6**	**-0.6**	**-0.7**	**-0.7**	**-0.7**	**-0.7**		**-0.7**	**-0.7**	-0.7
Endogenous debt dynamics 2/	**-3.6**	**0.4**	**-0.3**			**-0.8**	**-0.7**	**-0.6**	**-0.6**	**-0.6**	**-0.6**		**-0.6**	**-0.7**	
Contribution from nominal interest rate	0.2	0.2	0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.2	
Contribution from real GDP growth	-0.7	-1.0	-0.9			-1.0	-0.9	-0.8	-0.8	-0.8	-0.8		-0.8	-0.8	
Contribution from price and exchange rate changes	-3.1	1.1	0.4			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-0.6**	**8.0**	**1.6**			**5.0**	**4.2**	**3.2**	**2.3**	**1.0**	**0.1**		**1.4**	**0.5**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	14.2			13.6	13.0	12.4	12.1	11.8	11.7		11.3	10.6	
In percent of exports	...	...	132.7			122.5	116.1	111.4	108.3	105.5	103.8		102.1	94.2	
PV of PPG external debt	**...**	**...**	**14.1**			**13.6**	**13.0**	**12.4**	**12.1**	**11.8**	**11.7**		**11.3**	**10.6**	
In percent of exports	**...**	**...**	**131.4**			**122.5**	**116.1**	**111.4**	**108.3**	**105.5**	**103.8**		**102.1**	**94.2**	
In percent of government revenues	**...**	**...**	**80.5**			**74.3**	**69.2**	**65.4**	**62.8**	**60.7**	**59.5**		**56.7**	**50.7**	
Debt service-to-exports ratio (in percent)	**11.1**	**10.6**	**9.5**			**9.4**	**9.3**	**9.0**	**8.6**	**8.1**	**7.7**		**6.7**	**6.2**	
PPG debt service-to-exports ratio (in percent)	**11.1**	**10.6**	**9.5**			**9.4**	**9.3**	**9.0**	**8.6**	**8.1**	**7.7**		**6.7**	**6.2**	
PPG debt service-to-revenue ratio (in percent)	**6.9**	**6.7**	**5.8**			**5.7**	**5.6**	**5.3**	**5.0**	**4.7**	**4.4**		**3.7**	**3.3**	
Total gross financing need (Billions of U.S. dollars)	0.6	-0.6	-0.2			-0.7	-0.6	-0.4	-0.2	0.1	0.4		0.0	0.9	
Non-interest current account deficit that stabilizes debt ratio	4.7	-7.8	-0.8			-3.6	-2.9	-1.9	-1.0	0.3	1.2		0.0	0.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.4	4.8	3.9	4.2	0.9	4.8	5.0	4.7	4.5	4.5	4.5	4.7	4.5	4.5	4.5
GDP deflator in US dollar terms (change in percent)	14.9	-5.4	-1.7	7.5	8.1	-4.3	0.2	3.2	2.7	2.8	2.1	1.1	2.0	-2.8	1.4
Effective interest rate (percent) 5/	0.8	0.9	0.9	0.9	0.0	1.0	1.0	1.1	1.2	1.2	1.2	1.1	1.0	1.0	1.0
Growth of exports of G&S (US dollar terms, in percent)	10.8	11.7	8.4	7.5	8.0	4.1	5.8	7.6	7.4	7.8	7.3	6.7	6.3	3.5	6.0
Growth of imports of G&S (US dollar terms, in percent)	7.9	0.6	14.0	14.9	9.0	10.9	11.4	11.2	10.2	9.3	8.7	10.3	6.5	5.5	6.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	54.2	53.7	54.5	54.8	53.5	53.9	54.1	54.6	54.6	54.6
Government revenues (excluding grants, in percent of GDP)	14.4	16.0	17.5			18.3	18.7	18.9	19.2	19.4	19.6		20.0	21.0	20.2
Aid flows (in Billions of US dollars) 7/	1.0	0.7	0.5			0.6	0.7	0.8	0.8	0.8	0.9		1.2	2.1	
of which: Grants	0.6	0.5	0.3			0.5	0.5	0.6	0.6	0.6	0.7		0.9	1.3	
of which: Concessional loans	0.3	0.2	0.1			0.1	0.1	0.2	0.2	0.2	0.2		0.3	0.8	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.4	3.4	3.6	3.6	3.5	3.6		3.4	2.8	3.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			86.4	82.7	81.7	79.8	79.9	79.1		80.1	82.7	81.1
Memorandum items:															
Nominal GDP (Billions of US dollars)	19.0	18.9	19.2			19.3	20.3	21.9	23.6	25.3	27.0		37.2	64.0	
Nominal dollar GDP growth	18.8	-0.8	2.1			0.3	5.2	8.0	7.4	7.5	6.7	5.8	6.6	1.5	5.9
PV of PPG external debt (in Billions of US dollars)			2.5			2.5	2.6	2.7	2.8	2.9	3.1		4.2	6.7	
(PVt-PVt-1)/GDPt-1 (in percent)						0.1	0.3	0.4	0.6	0.6	0.7	0.4	0.7	0.4	0.6
Gross workers' remittances (Billions of US dollars)	3.5	4.4	4.9			5.7	6.3	6.9	7.4	7.8	8.3		11.1	19.9	
PV of PPG external debt (in percent of GDP + remittances)	...	...	11.2			10.5	9.9	9.4	9.2	9.0	8.9		8.7	8.1	
PV of PPG external debt (in percent of exports + remittances)	...	...	38.7			33.6	30.7	29.2	28.4	28.0	27.8		27.7	25.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.8			2.6	2.5	2.4	2.3	2.2	2.1		1.8	1.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2b. Nepal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to-GDP+remittances ratio								
Baseline	11	10	9	9	9	9	**9**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	11	10	10	9	8	6	**3**	-5
A2. New public sector loans on less favorable terms in 2014-2034 2/	11	10	10	10	10	11	**12**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	11	10	9	9	9	9	**9**	8
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	11	10	10	10	10	10	**9**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	11	10	10	9	9	9	**9**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	11	19	26	22	22	22	**19**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	11	17	24	21	20	20	**18**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	11	13	12	12	11	11	**11**	10
PV of debt-to-exports+remittances ratio								
Baseline	34	31	29	28	28	28	**28**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	34	33	32	30	27	22	**13**	-24
A2. New public sector loans on less favorable terms in 2014-2034 2/	34	31	31	32	32	34	**39**	44
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	34	30	29	28	28	27	**27**	25
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	34	31	32	31	31	31	**30**	26
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	34	30	29	28	28	27	**27**	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	34	83	114	70	68	67	**61**	40
B5. Combination of B1-B4 using one-half standard deviation shocks	34	73	101	66	65	64	**58**	39
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	34	30	29	28	28	27	**27**	25
PV of debt-to-revenue ratio								
Baseline	74	69	65	63	61	60	**57**	51
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	74	70	66	60	51	40	**19**	-25
A2. New public sector loans on less favorable terms in 2014-2034 2/	74	70	69	70	70	72	**80**	89
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	74	69	66	64	61	60	**57**	51
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	74	70	70	67	65	64	**60**	52
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	74	68	67	65	62	61	**58**	52
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	74	117	160	154	148	144	**124**	80
B5. Combination of B1-B4 using one-half standard deviation shocks	74	108	146	140	135	131	**114**	75
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	74	97	92	88	85	84	**80**	71

Table 2b. Nepal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034 (Continued)

(In percent)

Debt service-to-exports+remittances ratio								
Baseline	3	2	2	2	2	2	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	3	2	2	2	2	2	**2**	0
A2. New public sector loans on less favorable terms in 2014-2034 2/	3	2	2	2	2	2	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	3	2	2	2	2	2	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	3	3	2	2	2	2	**2**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	3	2	2	2	2	2	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	3	4	4	3	3	3	**4**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	4	3	3	3	**4**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	3	2	2	2	2	2	**2**	2
Debt service-to-revenue ratio								
Baseline	6	6	5	5	5	4	**4**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	6	5	5	4	4	4	**2**	0
A2. New public sector loans on less favorable terms in 2014-2034 2/	6	6	5	5	5	5	**5**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	6	6	5	5	5	5	**4**	3
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	6	6	5	5	5	5	**4**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	6	6	6	5	5	5	**4**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	6	6	6	7	6	6	**9**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	6	5	6	6	6	6	**8**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	6	8	8	7	7	6	**5**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	50	50	50	50	50	50	**50**	50

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.